

ATAR course Year 12 Physics

Student Name: _____

Practice Exam 2 (Units 3 & 4)

Time allowed

Reading time before commencing work: 10 minutes

Working time: 3 hours

Section	Number of questions	Number of questions to be answered	Suggested working time (minutes)	Number of marks	Percentage of exam
Section one: Short answer	13	13	50	52	28%
Section two: Problem-solving	7	6	90	91	50%
Section three: Comprehension	2	2	40	40	22%
Totals	22	21	180	183	100%

Notes to students

- Write your name in the space above.
- A Formulae and Data Booklet has been provided.
- The following items are approved for use in the examinations:
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters.
Special items: non-programmable calculators approved for use in the WACE examinations, drawing templates, drawing compass and a protractor
- You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

Disclaimer:

This is a practice examination. It represents Pearson Australia's view only of what would be useful preparation material for the externally assessed examination.

Section one: Short answer

30% (52 marks)

This section has **13** questions. Answer **all** questions. Write your answer in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Give final answers to **three** significant figures and include appropriate units where applicable.

When estimating numerical answers, show your working or reasoning clearly. Give final answers to **two** significant figures and include appropriate units where applicable.

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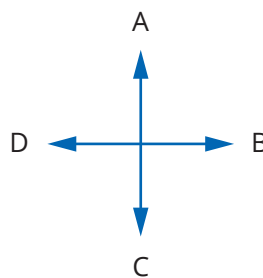
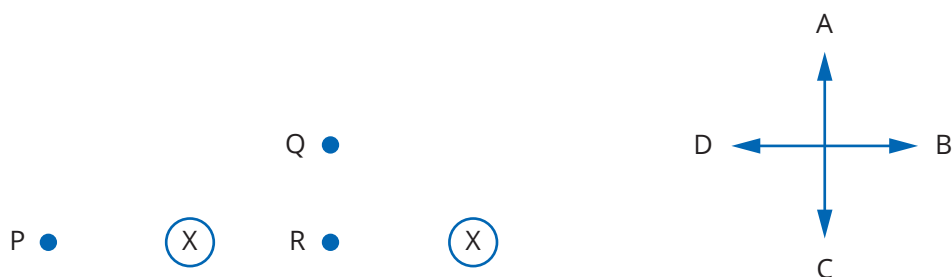
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Suggested working time: 50 minutes

Question 1

(3 marks)

The diagram represents two conductors, both perpendicular to the page and both carrying equal currents into the page. In these questions ignore any contribution from the Earth's magnetic field. The arrows A to D and letters E to G represent options from which you are to choose in answering the following questions.



E Out of page

F Into page

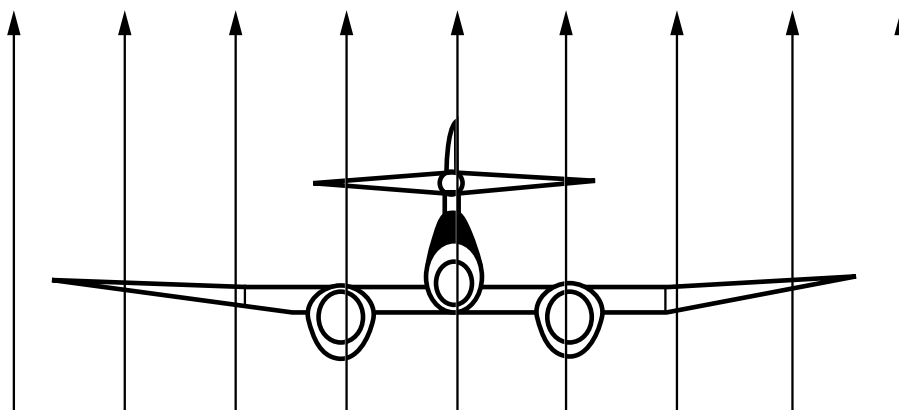
G Zero field

What is the direction of the magnetic field due to the two currents at the following points?

- a** point P ☐
- b** point Q ☐
- c** point R ☐

Question 2**(4 marks)**

The figure below shows an aircraft flying at right angles to the Earth's magnetic field ($B = 2.0 \times 10^{-6} \text{ T}$) at a speed of 160 m s^{-1} . The length of the wings from tip to tip is 30 m .



- a** Determine the magnitude of the EMF generated across the tips of the aircraft's wings. (2 marks)

- b** On the diagram above, place a '+' sign at the end of the wing that will acquire a positive charge and a '-' sign at the end of the wing that will acquire a negative charge. Explain your answer in the space provided. (2 marks)

Question 3**(5 marks)**

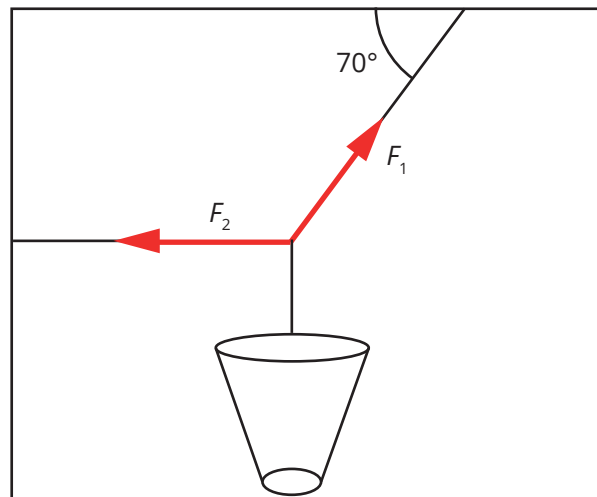
To determine the electrical permittivity, ϵ , in a newly developed material, a student measured the force of repulsion between two charged spheres separated by 1.0 cm . When each sphere carried a charge of $1.0 \mu\text{C}$, the force measured was 14.8 N .

- a** What is the electrical permittivity of this newly developed material? (3 marks)

- b** Show how this permittivity compares to the permittivity of a vacuum. (2 marks)

Question 4**(4 marks)**

Determine the tension in the two cords used to secure the 75 kg pot plant as represented in the diagram below.



Question 5**(4 marks)**

A domestic microwave oven is rated at 1000 W and uses a frequency of 2500 MHz. Calculate the number of photons emitted per second by the magnetron that produces the microwaves in the oven.

Question 6**(5 marks)**

A remote vehicle sits on an asteroid of diameter 28 km. On top of the remote vehicle is an object, at a height of 1.2 m from the surface of the asteroid. The remote vehicle propels the object vertically upwards at 0.5 m s^{-1} . The object returns to the asteroid's surface 208 seconds after launch.

- a** What is the acceleration due to gravity on the surface of the asteroid? (3 marks)

- b** What is the mass of the asteroid? (2 marks)

Question 7**(4 marks)**

An asteroid orbits the sun with a period of approximately 410 days. What is the average distance of the asteroid from the sun?

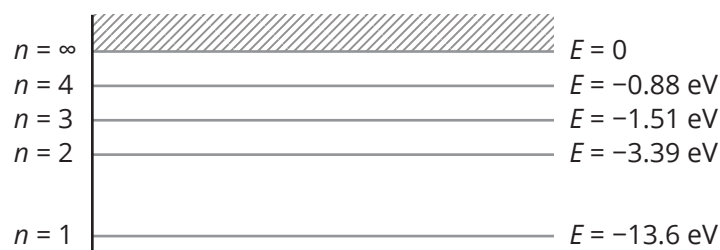
Question 8**(4 marks)**

An alternating generator with a rotor turning at a frequency of 50 Hz has 200 turns in its stator armature coils. Each coil has an area of 100 cm^2 . The magnetic field strength produced by the alternator's magnetising current is 0.50 T.

Determine the peak voltage output of the generator.

Question 9**(4 marks)**

Consider the energy level diagram for the hydrogen atom shown below. A photon of energy 14.0 eV collided with a hydrogen atom in the ground state.



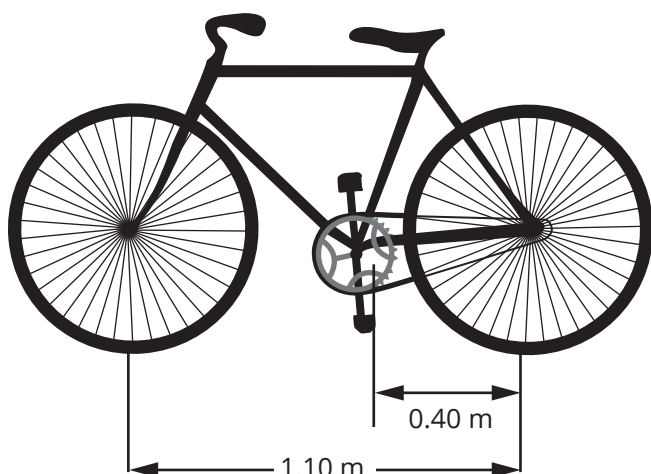
- a** Explain why this collision will eject an electron from the atom. (2 marks)

- b** Calculate the energy of the ejected electron in electron volts and joules. (2 marks)

Question 10**(4 marks)**

Rodney wants to determine the mass of his bicycle but the only equipment he has to weigh objects with is a set of bathroom scales. He has been able to determine that the centre of mass of the bicycle is 40 cm from the axis of the back wheel, and the distance between the wheel axles is 1.10 m.

Explain how Rodney can find the mass of his bike using only the bathroom scales. You may use a diagram to assist your explanation and indicate any mathematical relationships used to determine the bike's mass.



Question 11**(3 marks)**

Before the development of the Standard Model, it was thought that forces were exerted on particles by fields. In the Standard Model, it is believed that forces are exerted by the exchange of other particles.

- a** What name is given to the particles that give rise to forces? (1 mark)

- b** When two electrons approach each other, each experiences a force of repulsion from the other. How does the Standard Model explain this repulsion? (2 marks)

Question 12**(4 marks)**

The rest length of a spacecraft is 75 m. As it passes by a space station, its length is measured to be 60 m. What is the speed of the spacecraft relative to an observer on the space station?

Question 13**(4 marks)**

The absorption spectra for distant stars are red shifted. When this was first observed, it was interpreted as a Doppler shift.

- a** Explain what is meant by a Doppler shift in the context of a red shift of light. (2 marks)

It was later realised that the red shift of light from distant stars provides evidence to support the idea that the universe is expanding.

- b** Explain how the expansion of the universe accounts for the red shift of light from distant stars. (2 marks)

Section two: Problem-solving

50% (91 marks)

This section has 7 questions. Answer **all** questions. Write your answers in the spaces provided.

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Suggested working time: 90 minutes.

Question 14**(14 marks)**

A motorcycle stunt rider is designing a stunt to jump a series of trucks lined up side-by-side to give a total length of 50 m. The rider will need to reach a maximum height of 8 m to ensure that the trucks are cleared. The rider will take off from a ramp at an angle of 30° . The diagram below illustrates the jump.



- a** Calculate the speed at which the motorcycle needs to leave the ramp to reach the required height. (5 marks)

- b** What is this speed in kilometres per hour? (1 mark)

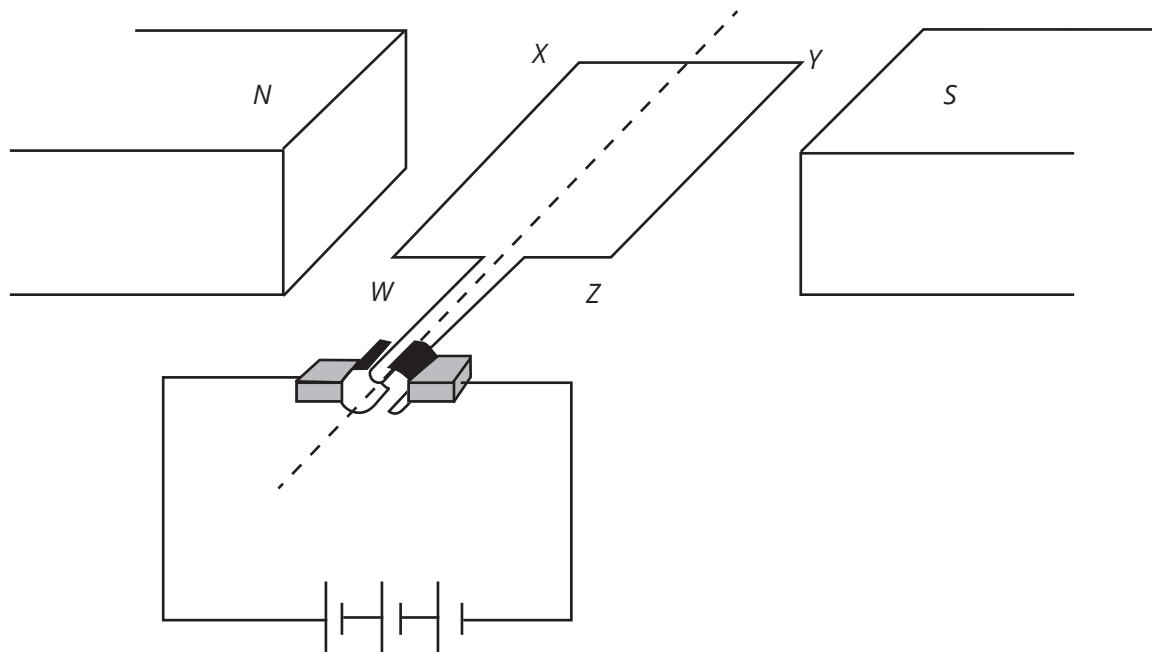
- c** Confirm that this take-off speed is sufficient to cover the required distance. (5 marks)

- d** The stunt rider plans to start the ride on the flat and accelerate towards the take-off ramp. Without changing the motorcycle or the take-off ramp, suggest what change to the design of the stunt can help ensure the rider takes off at the required speed. Explain your answer using the appropriate physics. (3 marks)

Question 15**(9 marks)**

The figure below shows a simplified diagram of a DC motor made by a physics student. It was built using two permanent magnets, a 6.0 V battery, a split ring commutator and a rectangular coil of wire consisting of 50 loops. As shown in the diagram, the coil is parallel to the magnetic field.

The resistance of the coil is $4.0\ \Omega$. The length of side WX is 6.0 cm and the length of side XY is 4.0 cm. The magnetic field is 0.08 T.



- a** Calculate the current flowing through the coil of wire. (2 marks)

- b** Determine the size of the force acting on side WX when the coil is in the position shown in the diagram. (2 marks)

- c** What is the size of the force acting on side XY when the coil is in the position shown in the diagram? Support your answer with suitable reasoning. (2 marks)

- d** State the role of the split ring commutator in this circuit and explain how it achieves its purpose. (3 marks)

Question 16**(16 marks)**

Electricity from a water-driven alternator of output 1000 V AC is transmitted by several kilometres of transmission lines to a step-down transformer which provides 250 V AC for a mill.

- a** Briefly describe the principle by which an alternator/generator produces electricity. Support your answer with a diagram. (4 marks)

- b** The primary (high-voltage) winding of the transformer consists of 2000 turns. How many turns are there in the secondary winding? (2 marks)

The alternator produces 200 kW of electrical power when the mill is working. The total resistance of the high-voltage transmission lines is $0.2\ \Omega$.

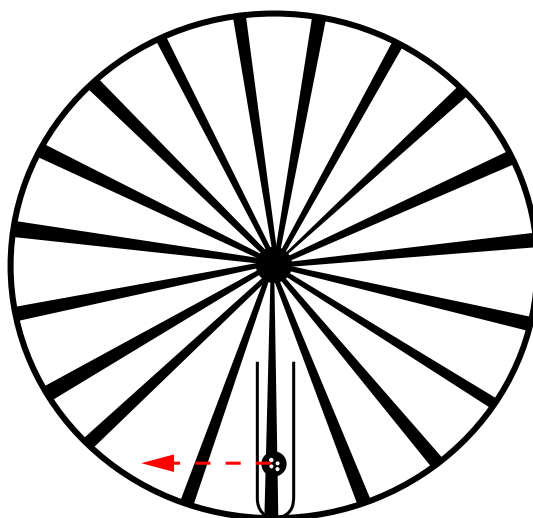
- c** What is the current flowing in the transmission lines when the mill is working? (3 marks)

- d** What is the power loss in the high-voltage transmission lines when the mill is working? (3 marks)

- e** The mill engineer is concerned about the power loss in the transmission lines from the alternator to the mill. State what steps the engineer could take to reduce this loss. Explain how this can reduce the transmission losses. (4 marks)

Question 17**(12 marks)**

A centrifuge is a device that rotates test tubes at very high speeds to separate components suspended in a liquid. The diagram below represents a centrifuge containing a test tube.



The centrifuge rotates at 75 000 rpm (revolutions per minute) in a clockwise direction. The open end of the test tube is 5 cm from the axis of rotation and the closed end is 15 cm from the axis.

- a** At any point in its motion, a suspended particle has the tendency to move in a straight line as represented in the diagram above. Explain why it does not continue to move in a straight line but rotates around the axis of the centrifuge. (2 marks)

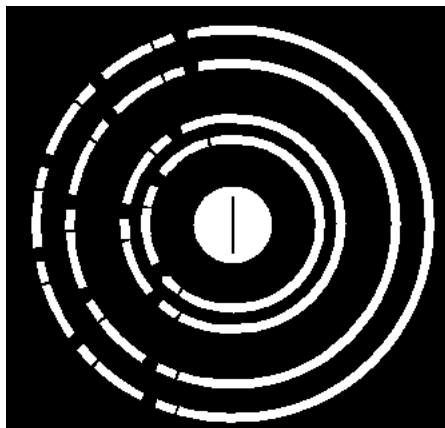
- b** Determine the centripetal acceleration in terms of 'g' (acceleration due to gravity) at the open end of the test tube. (6 marks)

- c** What is the centripetal acceleration at the closed end of the test tube? (2 marks)

- d** If the contents of the test tube have a mass of 11.5 g, what force must the bottom of the test tube withstand? (2 marks)

Question 18**(12 marks)**

The figure below shows a photograph of the diffraction patterns obtained by a group of students. They investigated the diffraction pattern created when a beam of electrons passed through a very thin film of gold foil and then the pattern created when X-rays were used instead of electrons.



The electrons used in the experiment were accelerated to a speed of $1.8 \times 10^7 \text{ m s}^{-1}$ before passing through the gold foil.

- a** Calculate the de Broglie wavelength of the electrons used. Show your working. (2 marks)

- b** State and explain the effect on the electron diffraction pattern if the accelerating voltage applied to the electron were increased. (3 marks)

- c** Determine the energy of a photon of the X-rays used. Show your working. (3 marks)

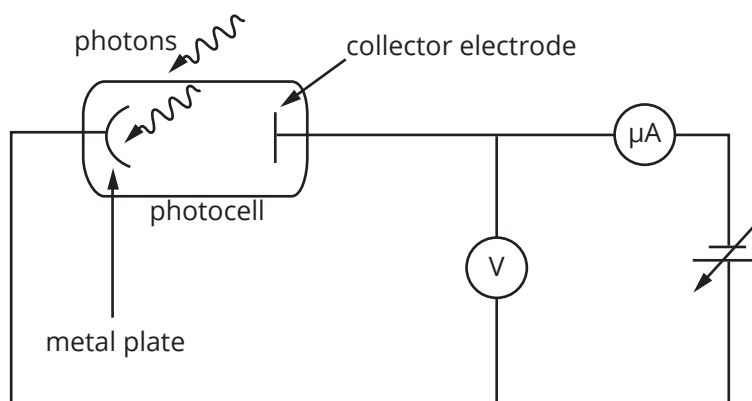
- d** State why the diffraction pattern obtained using electrons is the same as that obtained using X-rays. (2 marks)

- e Observations of the diffraction patterns for the interaction of X-rays and other electromagnetic radiation, with matter or through double slits (as in Young's experiment), support the wave model of light. However, we also observe properties of light that suggest it has a particle-like nature. Give two pieces of evidence that support the particle model of light. (2 marks)

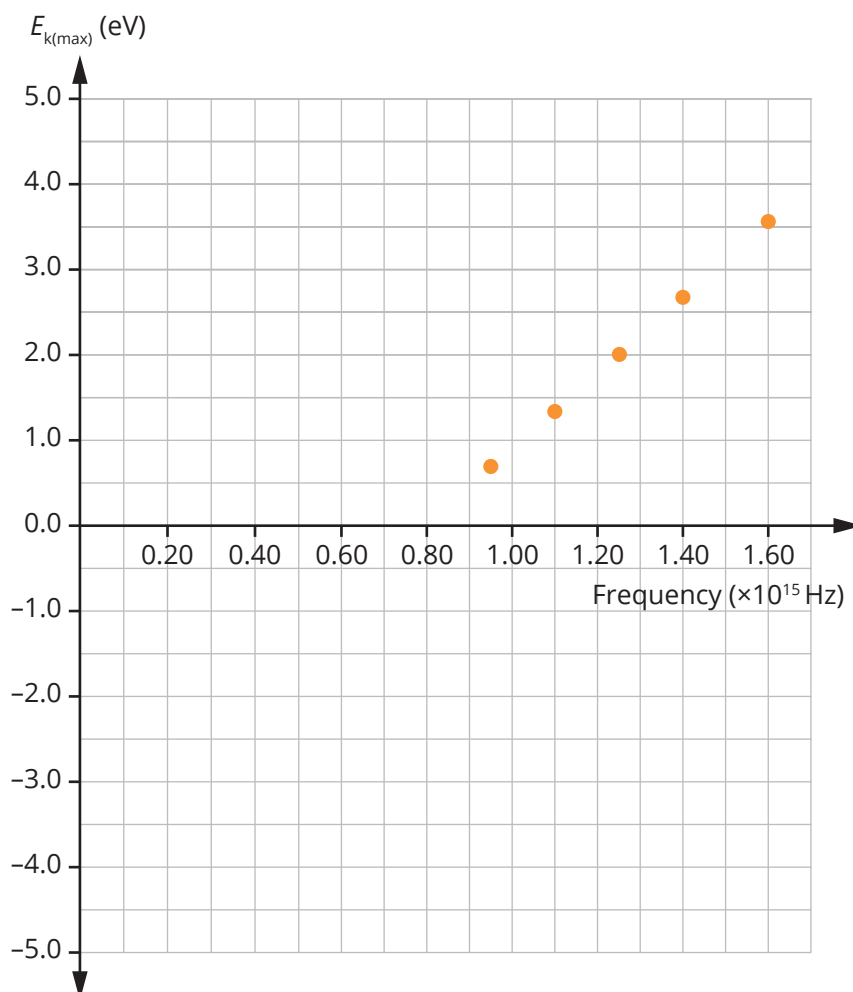
Question 19

(16 marks)

In an experiment designed to investigate the photoelectric effect, a group of physics students allowed light of various frequencies to fall on a metal plate coated in scandium (Sc) inside a photocell. The figure below shows the electric circuit used by the students to collect their data.



The data they collected is shown in the graph below.



- a** From the data on the graph, what value would the students obtain for Planck's constant? Include the units in your answer. (4 marks)

- b** From the data on the graph, what is the minimum energy required to remove photoelectrons from the scandium (Sc) plate? (2 marks)

- c** Explain why the existence of a threshold frequency supports the particle model and not the wave model for light. (2 marks)

The students replaced their scandium (Sc) photocell with one in which the metal plate was coated in caesium (Cs). Caesium has a work function of 2.1 eV.

The students then shone light from a 5.0 mW (5.0×10^{-3} W) monochromatic blue laser of wavelength $\lambda = 460$ nm onto the caesium photocell.

- d** Determine the velocity of the electrons ejected from the caesium. (6 marks)

The students repeated their original experiment using the caesium (Cs) photocell instead of a scandium (Sc) photocell.

- e** On the graph for the original experiment, draw a line for the graph that would be obtained for the caesium experiment. (2 marks)

Question 20

(12 marks)

Quarks are one of the two elementary particles in the Standard Model of particle physics. There are six types of quarks. Each quark has its own antiquark.

Table of quarks			
Name	Quark symbol	Quark electrostatic charge	Antiquark symbol
Up	u	$+\frac{2}{3}e$	$\bar{u}$
Down	d	$-\frac{1}{3}e$	$\bar{d}$
Strange	s	$-\frac{1}{3}e$	$\bar{s}$
Charmed	c	$+\frac{2}{3}e$	$\bar{c}$
Bottom	b	$-\frac{1}{3}e$	$\bar{b}$
Top	t	$+\frac{2}{3}e$	$\bar{t}$

Baryons are made up of three quarks. The lambda baryon, Λ^0 , is composed of an up, down and strange quark. It has a rest mass of 1.98×10^{-27} kg and mean lifetime of 2.63×10^{-10} s.

- a** Show that Λ^0 baryon is neutral. (1 mark)

- b** How much energy will be released if a Λ^0 baryon at rest decays to electromagnetic radiation? (2 marks)

- c** How much energy will be released if a Λ^0 baryon travelling at $0.95c$ decays to electromagnetic radiation? (3 marks)

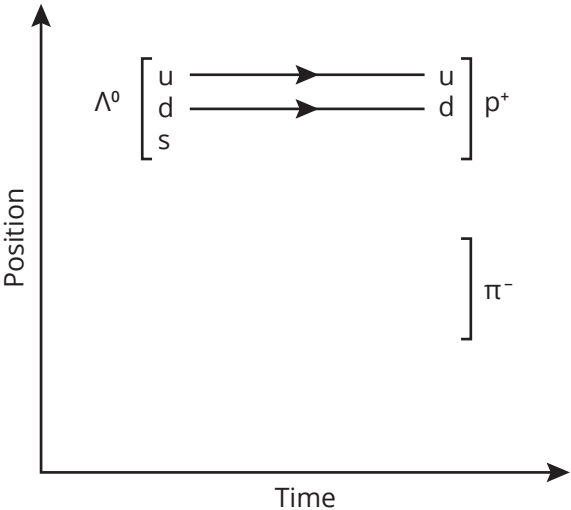
- d** What will be the mean lifetime of a Λ^0 baryon when measured in the laboratory if it is travelling at $0.95c$? (3 marks)

The Λ^0 baryon typically decays to a proton and π^- meson antiparticle, as shown below. This decay is mediated by a W^- boson.

$$\Lambda^0 \rightarrow p^+ + \pi^-$$

Table of particles for decay of Λ^0	
Particle/antiparticle symbol	Quark composition
Λ^0	u d s
p^+	u d u
π^-	d $\bar{u}$

e Complete the Feynman diagram below to represent this decay of the Λ^0 baryon. (3 marks)



Section three: Comprehension

20% (40 marks)

This section has 2 questions. You must answer **both** questions. Write your answers in the spaces provided.

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Suggested working time: 40 minutes.

Question 21

(19 marks)

Audio induction loops



This symbol is becoming increasingly common in places such as railway platforms, concert theatres, cinemas and seminar rooms. It is the symbol for a piece of technology that assists hearing-impaired people to hear sounds more clearly. The technology is known as an audio induction loop system, also called audio-frequency induction loops or more simply, hearing loops.

Figure 1: Standard symbol to show that a facility has an induction hearing loop installed

A hearing loop consists of a physical wire loop or an array of looped wires placed around an area such as a room or a building. The loop creates a magnetic field around the looped space, which can be picked up by a hearing aid with a telecoil, Cochlear Implant (CI) processors, and specialised hand-held hearing loop receivers for individuals without a telecoil-compatible hearing aid.

The telecoil was originally developed to be fitted into hearing aids to assist hearing-impaired people hear over the telephone. The copper coil picks up electromagnetic fields from coils within a telephone. The telecoil then converts those to an audio signal. Figure 2 shows a representation of a telecoil.

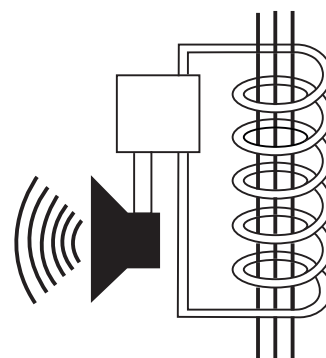
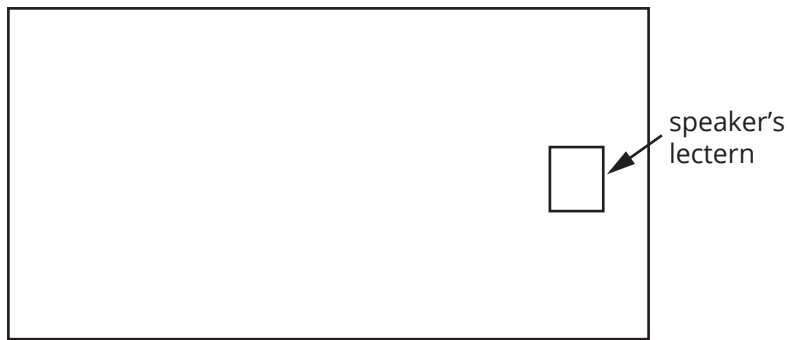


Figure 2: Representation of a telecoil

An audio induction loop system usually consists of a loop of wire around the room (or the space concerned) connected to an amplifier. An audio signal from a microphone, television, stereo or some other source goes to the amplifier, which sends a current through the loop of wire. As the current from the amplifier flows through the loop, it creates a magnetic field within the looped area. This magnetic field is detected by the telecoil, which in turn converts the signal back to an audio signal. In the hearing aid, the telecoil is attached to a loudspeaker that converts the electrical signal to sound. A loudspeaker also works on the principle of induction, using the current to make a diaphragm vibrate, via a magnet, to create sound waves.

The success of an induction hearing loop in a seminar room requires that the speaker's voice be picked up by a microphone. Microphones also work on the principle of induction to convert the sound to an electrical signal and are, in fact, essentially the reverse of a loudspeaker.

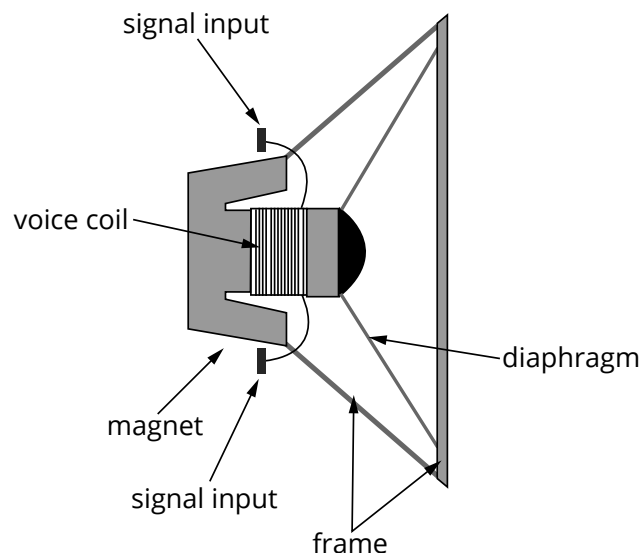
- a** The rectangle below represents the top view of a seminar room. The front of the room is indicated by a speaker's lectern. Draw a schematic diagram to represent the room fitted with an audio induction loop system. Label the parts of the system required for the loop to operate. (3 marks)



- b** What type of current is needed for the system to operate? Explain your answer. (2 marks)

- c** Identify two ways the size of the induced current in the telecoil may be increased. Give the relationship that supports your answer. (3 marks)

- d** The telecoil converts the signal to sound using a speaker. The diagram below represents a speaker. Explain how a speaker converts the electrical signal to sound. Your answer needs to indicate how the frequency of the sound is produced correctly by the speaker. (5 marks)



- e** Explain how a microphone converts sound to an electrical signal. Support your explanation with a labelled diagram. (6 marks)

Vision

Our capacity to detect light is controlled by four types of light receptors, or photoreceptors. Rods allow us to detect black, white or shades of grey, and the three types of cones are responsible for colour vision. Light is perceived when pigments in the rods and cones capture photons, and it has been shown that they can respond to a single photon.

Rods are able to function at low levels of light and are responsible for night, or scotopic, vision. This sensitivity to light is a result of several rods converging on a single nerve fibre. Rods contain the pigment protein rhodopsin, which undergoes chemical changes on absorption of photons that lead to an electrical signal being sent along the optic nerve to the brain. Rhodopsin absorbs over the wavelengths of approximately 400 to 600 nm, with its maximum absorbance at about 510 nm (Figure 1).

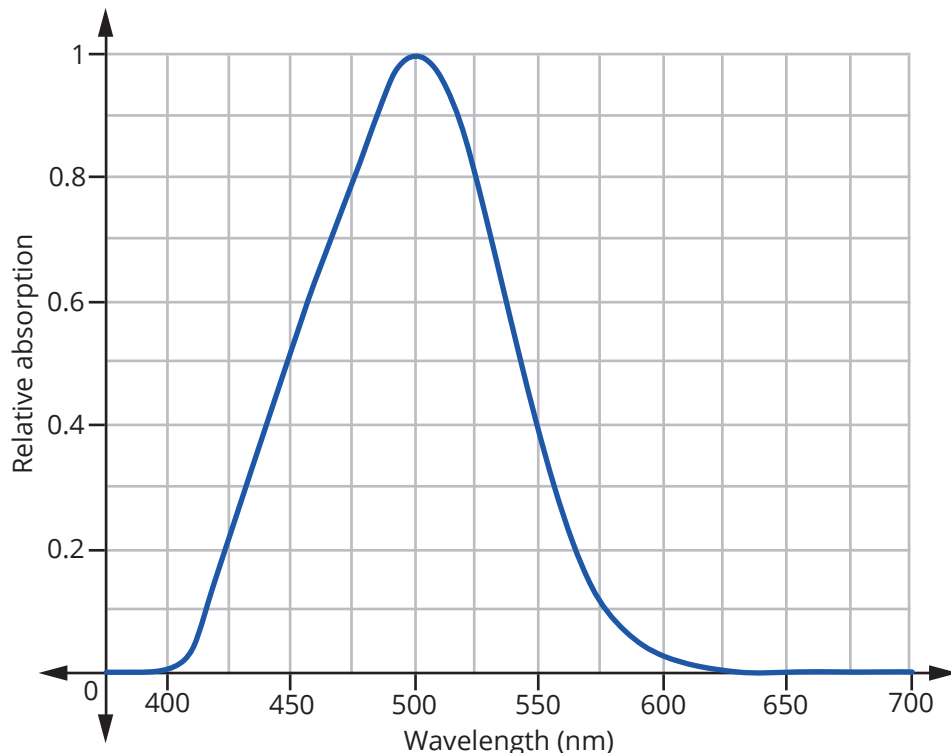


Figure 1: Absorption spectrum for rhodopsin

Humans have three types of cones: L-type cones that are most sensitive to longer wavelength light (red end of the spectrum); M-type cones responding most to medium-wavelength light (green region of the spectrum); and S-type cones detecting short-wavelength light (blue end of the spectrum). The pigments found in cones, opsins, are similar to rhodopsin. On absorbing light, they undergo conformational changes that initiate a series of reactions to produce an electrochemical signal in the optic nerve. The L-type cones are most sensitive and the S-type cones the least sensitive (Figure 2).

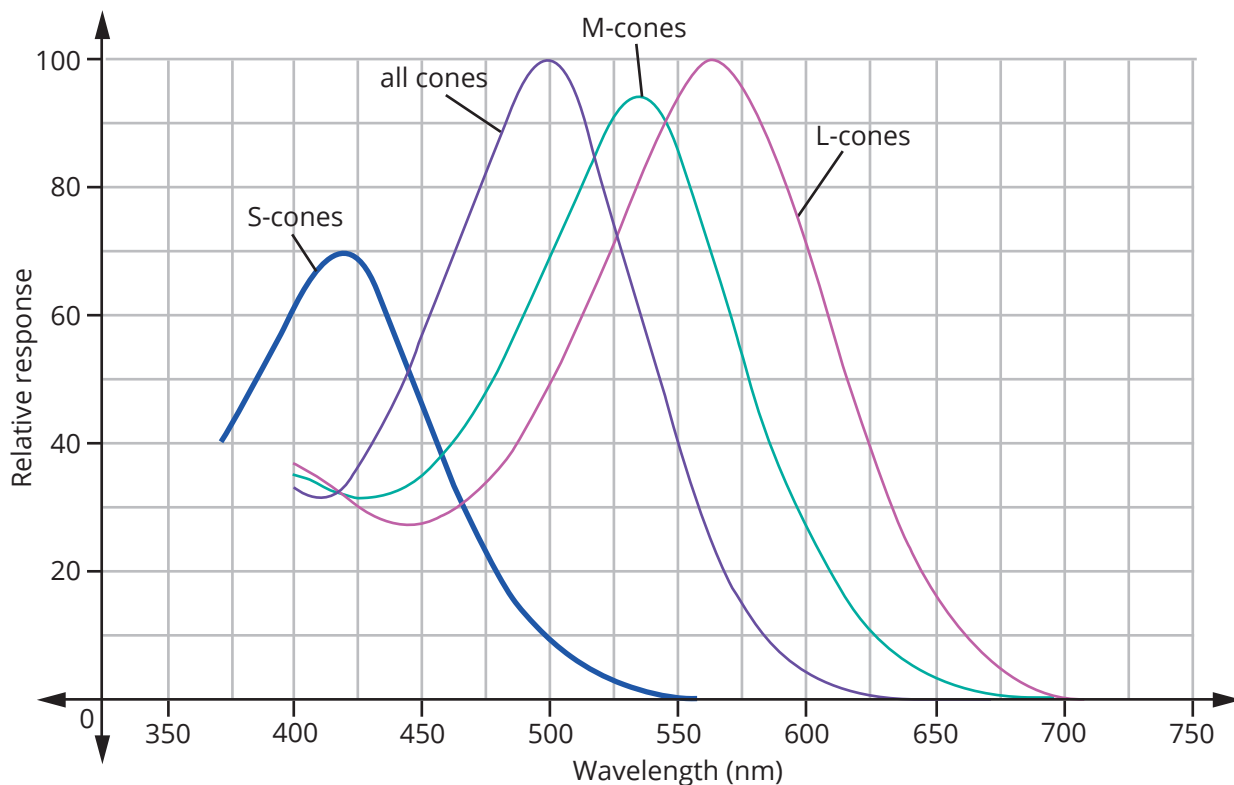


Figure 2: Absorption spectra for cone pigments

Our perception of colour results from the combined responses of cones. If light of wavelength 390 nm enters the eye, only blue light will be perceived. If light wavelength 570 nm enters the eye, M-type and L-type cones will be stimulated and a yellow shade of colour will be perceived.

How bright we perceive light to be depends on the intensity of the light, or the number of photons incident per second. However, not all photons that strike the retina are absorbed by the pigments in the rods and cones—some are transmitted through the retina. In general, the more photons absorbed by the pigments in a given timeframe, the greater the stimulus to the brain. The absorbance of light is wavelength dependant (Figures 1 and 2).

As well as the rods and cones, the lens and cornea have role in the wavelengths of light which can be sensed by our eyes. The lens does not allow wavelengths shorter than about 380 nm to pass through it, and the cornea is opaque to wavelengths less than 300 nm.

- a** What is meant by the term 'scotopic vision'? (1 mark)

- b** It is stated that rods and cones can respond to a single photon of light. What is the energy associated with a single photon of light of wavelength 550 nm? (2 marks)

- c** At what frequency are rods most sensitive? To which part of the electromagnetic spectrum does this frequency correspond? (3 marks)

- d** The article indicates that one factor affecting our perception of brightness is the intensity of the light. What is meant by intensity of light? (1 mark)

- e** For light of the same intensity, compare the perception of light of wavelength 510 nm to light of wavelength 460 nm at night. Explain your answer. (4 marks)

- f** What is the approximate maximum wavelength that can be perceived by human eyes? What colour would this wavelength be perceived as? (2 marks)

- g** What colour would be perceived, in daylight, when light of wavelength 620 nm is incident on the eye. Explain your answer, including an indication of the extent of any response by photoreceptors. (3 marks)

- h** If light of wavelength 620 nm with twice the intensity of that used in f above is incident on the eyes, how will its perception compare to the perception of the light in f? (2 marks)

- i** Contrast the way in which the pigments in the rods and cones respond to light to produce an electrical signal to the way that a metal will respond to exposure to light to produce an electrical signal. (3 marks)

This image shows a full page of blank, lined paper. It features approximately 28 horizontal blue or grey lines spaced evenly apart, typical of notebook paper. The lines extend across the entire width of the page, leaving small margins at the top and bottom. There are no vertical lines, text, or other markings on the page.

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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